

Vermont State Mathematics Coalition Talent Search Test #3

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March 23, 2025

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1 Problem #1

Problem Note that in chess, a bishop attacks all squares that share a diagonal with it. An 8 x 8 chessboard has its squares labeled using the integers 1 through 64 inclusive as shown here.

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64

- (a) If n is divisible by 2,3,5, or 7, show that there exists a $2n$ -digit integer N with nonzero digits such that no chop of N is divisible by n .
- (b) If N is a $2n$ -digit integer with nonzero digits, and n is not divisible by 2,3,5, or 7, show that there is some chop of N that is divisible by n .

(a) Solution

(b) Solution

2 Problem #2

2.1 a

Problem For an integer n , let $S(n)$ denote the sum of the digits of n and let $D(n)$ denote the total number of positive divisors of n . For $n = (10+1)(10^2+1)(10^4+1)\dots(10^{2^{2024}}+1) - 1$, compute the value of $D(S(D(D(S(n)))))$.

Solution

2.2 b

Problem Let A be the answer to part (a). Suppose x_1 and x_2 are the two real solutions to the equation $10 - 4^{A^x} = 4^{2-A^x}$, with $x_1 > x_2$. Compute $(\sqrt{A})^{x_1-x_2}$.

Solution

2.3 c

Problem Let B be the answer to part (b). Point P lies inside square $VMTS$ in such a way that the areas of PVM , PMT , and PTS are B , $3B$, and $9B$, respectively. Find the area of triangle PSV .

Solution

3 Problem #3

Problem Find the number of ordered pairs of nonnegative integers (a, b) such that $9a + 6\sqrt{ab} + b + 2025 = 702\sqrt{a} + 234\sqrt{b}$.

Solution Let $x, y = \sqrt{a}, \sqrt{b}$ so that $9x^2 + 6xy + y^2 + 2025 = 702x + 234y \implies (3x + y)^2 + 2025 = 234(3x + y)$. Manipulating we have a quadratic in $3x + y$:

$$(3x + y)^2 - 234(3x + y) + 2025 = 0$$

then we can apply the quadratic formula to find $3x + y = \frac{234 \pm \sqrt{234^2 - 4(1)(2025)}}{2} = 9, 225$. Then we're left with two cases. Either $3x + y = 9$ or $3x + y = 225$. If $3x + y = 9$, only four nonnegative integers x satisfy $3x + y = 9$ where y is also a nonnegative integer. Similarly, if $3x + y = 225$, it's clear that for y ranging from 0 to 225, only $(\frac{225}{3} = 75) + 1$ (for the case where $x = 0$) = 76 nonnegative integers x satisfy $3x + y = 225$. This gives $4 + 76 = \boxed{80}$ pairs $(x, y) = (\sqrt{a}, \sqrt{b})$, and since these x, y are nonnegative integers, it follows that the 80 corresponding $(a, b) = (x^2, y^2)$ are also nonnegative integer pairs satisfying $9a + 6\sqrt{ab} + b + 2025 = 702\sqrt{a} + 234\sqrt{b}$.

4 Problem #4

Problem A polynomial $p(x)$ of degree at most 2025 has the property that each of the values $p(0), p(1), \dots, p(2025)$ is either 0 or 1. Compute the maximum possible value of $p(2026)$.

Solution

5 Problem #5

Problem Triangle ABC has $AB = 13, AC = 14, BC = 15$. Squares $ABDE, BCHI, CAFG$ are constructed outside triangle ABC , forming a hexagon $DEFGHI$. Squares $DEKJ, EFML, FGON, GHQP, HISR, IDUT$ are constructed outside hexagon $DEFGHI$, forming a dodecagon $JKLMNOPQRSTU$. Find the area of $JKLMNOPQRSTU$.

Solution I solve this problem, in essence, by repeated use of the law of cosines. First, separate $JKLMNOPQRSTU$ into the polygons $AEKJDBC GONF, AEKLMNF, BDJUTSI$, and $CHRQPOG$.

$AEKJDBC GONF$. Note that the area of $ABC = \sqrt{s(s-AB)(s-BC)(s-CA)}$ where $s = \frac{AB+BC+CA}{2}$ by Heron's formula. So, $A_{ABC} = 84$. Then, outside ABC we have squares with areas known by construction: $A_{ABDE} = 13^2 = 169, A_{BCHI} = 15^2 = 225, A_{CAFG} = 14^2 = 196$. We know that the squares $DEKJ, FGON$, and $HISR$ have the same side lengths as the squares they were attached to. Therefore, $A_{DEKJ} = 13^2 = 169, A_{FGON} = 14^2 = 196, A_{HISR} = 15^2 = 225$. Now that we have all components of $AEKJDBC GONF$, we can find that $A_{AEKJDBC GONF} = A_{ABC} + A_{ABDE} + A_{BCHI} + A_{CAFG} + A_{DEKJ} + A_{FGON} + A_{HISR} = 84 + 169 + 225 + 196 + 169 + 196 + 225 = 1264$.

$AEKLMNF$. Next we find the area of $AEKLMNF$. First, by law of cosines $\angle BAC = \arccos\left(\frac{AC^2+AB^2-BC^2}{2(AC)(AB)}\right) \therefore \angle EAF = 360^\circ - 2(90^\circ) - \angle BAC = 180^\circ - \angle BAC$. By law of cosines we also find that $FE = \sqrt{EA^2 + FA^2 - 2(EA)(FA)\cos(\angle EAF)}$. With FE law of cosines tells us that $\angle EFA = \arccos\left(\frac{FE^2+FA^2-EA^2}{2(FE)(FA)}\right) \therefore \angle MFN = 360^\circ - 180^\circ - 90^\circ - \angle EFA = 90^\circ - \angle EFA$. Because $EFML$ is a square, $LE = MF = FE$. Then law of cosines says $MN = \sqrt{NF^2 + MF^2 - 2(NF)(MF)\cos(\angle MFN)}$. Then, since we have $\angle EFA$ and $\angle EAF$, we know that $\angle FEA = 180^\circ - \angle EAF - \angle EFA \implies \angle LEK = 360^\circ - 180^\circ - 90^\circ - \angle FEA$ because $\angle LEF$ is inside a square. With $\angle LEK$ we can, again by law of cosines, find $LK = \sqrt{KE^2 + LE^2 - 2(LE)(KE)\cos(\angle LEK)}$. We then have

$$A_{MFN} = \sqrt{s(s-MF)(s-NF)(s-MN)} \text{ where } s = \frac{MF + NF + MN}{2}$$

$$A_{EAF} = \sqrt{s(s-EA)(s-FA)(s-EF)} \text{ where } s = \frac{EA + FA + EF}{2}$$

$$A_{LEK} = \sqrt{s(s-LE)(s-KE)(s-LK)} \text{ where } s = \frac{LE + KE + LK}{2}$$

by Heron's formula and

$$A_{FELM} = FE^2$$

Evaluating all generalizations above with the initial lengths of ABC and lengths $NF = 14, FA = 14, EA = 13, KE = 13$ and summing the defined areas gives $A_{AEKLMNF} = 841.5$.

$BDJUTSI$ Next, we apply an identical procedure as described for $AEKLMNF$ to find $BDJUTSI$. First, by law of cosines $\angle CBA = \arccos\left(\frac{BA^2+BC^2-CA^2}{2(BA)(BC)}\right) \therefore \angle IBD = 360^\circ - 2(90^\circ) - \angle CBA = 180^\circ - \angle CBA$. By law of cosines we also find that $DI = \sqrt{IB^2 + DB^2 - 2(IB)(DB)\cos(\angle IBD)}$. With DI law of cosines tells us that $\angle IDB = \arccos\left(\frac{DI^2+DB^2-IB^2}{2(DI)(DB)}\right) \therefore \angle UDJ = 360^\circ - 180^\circ - 90^\circ - \angle IDB = 90^\circ - \angle IDB$. Because $IDUT$ is a square, $TI = UD = DI$. Then law of cosines says $UJ = \sqrt{JD^2 + UD^2 - 2(JD)(UD)\cos(\angle UDJ)}$. Then, since we have $\angle IDB$ and $\angle IBD$, we know that $\angle DIB = 180^\circ - \angle IBD - \angle IDB \implies \angle TIS = 360^\circ - 180^\circ - 90^\circ - \angle DIB$ because $\angle TID$ is inside a square. With $\angle TIS$ we can, again by law of cosines, find $TS = \sqrt{SI^2 + TI^2 - 2(SI)(TI)\cos(\angle TIS)}$. We then have

$$A_{UDJ} = \sqrt{s(s-UD)(s-JD)(s-UJ)} \text{ where } s = \frac{UD + JD + UJ}{2}$$

$$A_{IBD} = \sqrt{s(s-IB)(s-DB)(s-ID)} \text{ where } s = \frac{IB + DB + ID}{2}$$

$$A_{TIS} = \sqrt{s(s-TI)(s-SI)(s-TS)} \text{ where } s = \frac{TI + SI + TS}{2}$$

by Heron's formula and

$$A_{DITU} = DI^2$$

Evaluating all generalizations above with the initial given lengths of ABC and lengths $JD = 13, DB = 13, IB = 15, SI = 15$ and summing the defined areas gives $A_{BDJUTSI} = 972$.

CHRRQPOG Just like above. First, by law of cosines $\angle ACB = \arccos\left(\frac{CB^2+CA^2-AB^2}{2(CB)(CA)}\right) \therefore \angle GCH =$

$360^\circ - 2(90^\circ) - \angle ACB = 180^\circ - \angle ACB$. By law of cosines we also find that $HG = \sqrt{GC^2 + HC^2 - 2(GC)(HC)\cos(\angle GCH)}$

With HG law of cosines tells us that $\angle GHC = \arccos\left(\frac{HG^2+HC^2-GC^2}{2(HG)(HC)}\right) \therefore \angle QHR = 360^\circ - 180^\circ - 90^\circ - \angle GHC = 90^\circ - \angle GHC$. Because $GHQP$ is a square, $PG = QH = HG$. Then law of cosines says $QR = \sqrt{RH^2 + QH^2 - 2(RH)(QH)\cos(\angle QHR)}$. Then, since we have $\angle GHC$ and $\angle GCH$, we know that $\angle HGC = 180^\circ - \angle GCH - \angle GHC \implies \angle PGO = 360^\circ - 180^\circ - 90^\circ - \angle HGC$ because $\angle PGH$ is inside a square. With $\angle PGO$ we can, again by law of cosines, find $PO = \sqrt{OG^2 + PG^2 - 2(OG)(PG)\cos(\angle PGO)}$. We then have

$$A_{QHR} = \sqrt{s(s-QH)(s-RH)(s-QR)} \text{ where } s = \frac{QH + RH + QR}{2}$$

$$A_{GCH} = \sqrt{s(s-GC)(s-HC)(s-GH)} \text{ where } s = \frac{GC + HC + GH}{2}$$

$$A_{PGO} = \sqrt{s(s-PG)(s-OG)(s-PO)} \text{ where } s = \frac{PG + OG + PO}{2}$$

by Heron's formula and

$$A_{HGPQ} = HG^2$$

Evaluating all generalizations above with the initial given lengths of ABC and lengths $RH = 13, HC = 13, GC = 15, OG = 15$ and summing the defined areas gives $A_{CHRRQPOG} = 1093.5$.

Recall from our separations that $A_{JKLMNOPQRSTU} = A_{AEKJDBCGONF} + A_{AEKLMNF} + A_{BDJUTSI} + A_{CHRRQPOG} = 1264 + 841.5 + 972 + 1093.5 = \boxed{4,171}$.

6 Problem #6

Problem Let $g(n) = 2n - 1 - \lfloor [n\sqrt{2}] \sqrt{2} \rfloor$, where $\lfloor x \rfloor$ denotes the greatest integer $\leq x$. Prove that for all positive integers n ,

$$g(n)^2 + g(n+1)^2 + g(n+2)^2 + g(n+3)^2 + g(n+4)^2$$

equals either 1 or 2.

Proof